

VSFTPD LAB EXAM PRACTICE

Q1. Configure an FTP server using vsftpd

Install the FTP server package

Start and enable the FTP service

Verify the service is running

Confirm FTP is listening on port 21

Q2. Configure firewall for FTP access

Check if FTP service is available in firewall services

Allow FTP service permanently

Reload firewall configuration

Verify FTP service is allowed

Q3. Configure anonymous FTP access

Place a file named public.txt in the FTP public directory

Enable anonymous login in configuration

Restart FTP service

Verify anonymous users can access and download the file

Q4. Configure local user authentication

Create a local user ftpuser

Set password for the user

Enable local user login in configuration

Login using FTP client with this user

Verify access to user home directory

Q5. Perform file transfer operations

Download a file from FTP server to client

Upload a file from client to server

Upload multiple files from client

Download multiple files from server

Q6. Restrict user to home directory

Configure FTP to restrict users to their home directory

Restart FTP service

Verify user cannot access directories outside home

Q7. Disable anonymous access

Modify configuration to disable anonymous login

Restart FTP service

Verify anonymous login is denied

Q8. Configure custom upload directory

Create a directory for uploads

Set appropriate permissions for writing

Enable upload capability in FTP configuration

Upload files from client to this directory

Q9. Troubleshoot FTP connection issues

Verify FTP service status

Check if FTP port is active

Verify firewall configuration

Restart FTP service if required

Q10. Complete end to end FTP setup

Install and configure FTP server

Allow FTP in firewall

Create a user labuser

Perform login from client

Upload and download files successfully

Verify full working setup

```
#####  
# VSFTPD COMPLETE LAB GUIDE  
# Covers Q1-Q10 (Server + Client)  
# Each command includes:  
# WHY -> purpose  
# WHEN -> when to use  
# HOW -> what it does  
#####  
#####  
# ----- SERVER SIDE -----  
#####
```

```
echo "Q1: INSTALL FTP SERVER"
```

```
# Install vsftpd package  
# WHY: Provides FTP service daemon  
# WHEN: First step before configuration  
dnf install vsftpd -y
```

```
# Check service status  
# WHY: Verify service state  
systemctl status vsftpd
```

```
# Start service  
# WHY: Activate FTP server immediately  
systemctl start vsftpd
```

```
# Enable at boot  
# WHY: Auto start after reboot  
systemctl enable vsftpd
```

```
# Restart service
# WHY: Apply config changes
systemctl restart vsftpd
```

```
# Verify again
systemctl status vsftpd
```

```
echo "VERIFY PORT"
```

```
# Check if FTP is listening on port 21
# WHY: Confirms server is accepting connections
lsof -i:21
```

```
echo "Q2: FIREWALL CONFIGURATION"
```

```
# Check available services
# WHY: Confirm FTP service exists
firewall-cmd --get-services | grep ftp
```

```
# Allow FTP permanently
# WHY: Allow external clients to connect
firewall-cmd --add-service=ftp --permanent
```

```
# Reload firewall
# WHY: Apply rules
firewall-cmd --reload
```

```
# Verify firewall rules
```

```
firewall-cmd --list-all
```

```
echo "Q3: ANONYMOUS FTP SETUP"
```

```
# Check config file
```

```
# WHY: Verify settings
```

```
cat /etc/vsftpd/vsftpd.conf
```

```
# Enable anonymous login
```

```
# WHY: Allow users without account
```

```
# HOW: Edit file manually
```

```
# anonymous_enable=YES
```

```
# Create public file
```

```
# WHY: Test anonymous access
```

```
touch /var/ftp/pub/public.txt
```

```
# Restart service
```

```
systemctl restart vsftpd
```

```
echo "Q4: LOCAL USER AUTHENTICATION"
```

```
# Create user
```

```
# WHY: Allow login with credentials
```

```
useradd ftpUser
```

```
# Set password
passwd ftpUser
```

```
# Enable local login in config
# local_enable=YES
```

```
# Restart service
systemctl restart vsftpd
```

```
echo "Q6: CHROOT (RESTRICT USER)"
```

```
# WHY: Prevent user from accessing system directories
# Add in config:
# chroot_local_user=YES
```

```
systemctl restart vsftpd
```

```
echo "Q7: DISABLE ANONYMOUS"
```

```
# WHY: Increase security
# Set:
# anonymous_enable=NO
```

```
systemctl restart vsftpd
```

```
echo "Q8: CUSTOM UPLOAD DIRECTORY"
```

```
# Create directory
```

```
# WHY: Allow controlled uploads
```

```
mkdir /ftp
```

```
# Set permissions
```

```
# WHY: Allow writing
```

```
chmod 777 /ftp
```

```
# Enable uploads in config:
```

```
# write_enable=YES
```

```
systemctl restart vsftpd
```

```
echo "Q9: TROUBLESHOOTING"
```

```
# Check service
```

```
systemctl status vsftpd
```

```
# Check port
```

```
lsof -i:21
```

```
# Check firewall
```

```
firewall-cmd --list-all
```

```
# Restart if needed
```

```
systemctl restart vsftpd
```

```
#####  
# ----- CLIENT SIDE -----  
#####
```

```
echo "INSTALL FTP CLIENT"
```

```
# Install FTP client  
# WHY: Required to connect to server  
dnf install ftp -y
```

```
echo "CONNECT TO SERVER"
```

```
# Connect to FTP server  
# WHY: Start FTP session  
ftp <server-ip>
```

```
# Enter username (ftpUser or anonymous)  
# Enter password
```

```
echo "Q5: FILE OPERATIONS"
```

```
# Show files  
# WHY: List server directory
```



```
ls
```

```
# Print working directory  
# WHY: Know current location
```

```
pwd
```

```
# Change server directory
```

```
# WHY: Navigate server
```

```
cd /path
```

```
# Change local directory
```

```
# WHY: Navigate client system
```

```
lcd /path
```

```
echo "DOWNLOAD FILE"
```

```
# Download file from server
```

```
# WHY: Copy server file to client
```

```
get filename
```

```
echo "UPLOAD FILE"
```

```
# Upload file to server
```

```
# WHY: Send file from client
```

```
put filename
```

```
echo "MULTIPLE DOWNLOAD"
```

```
# Disable prompt
# WHY: Avoid confirmation for each file
prompt off

# Download all files
mget *

echo "MULTIPLE UPLOAD"

# Upload multiple files
mput *

echo "COMMON ERRORS"

# 550 Failed to open file
# WHY: File does not exist OR permission issue

# local: No such file
# WHY: File not present in client directory

# Passive mode refused
# WHY: Firewall or network issue

echo "EXIT"

# Close FTP session
```

bye

#####

Q10 FINAL VALIDATION

#####

SERVER:

Service running

systemctl status vsftpd

Port open

lsof -i:21

Firewall allowed

firewall-cmd --list-all

CLIENT:

Login works

Upload works

Download works

#####

END OF LAB

#####

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Q1. Configure an FTP server using vsftpd

Install the FTP server package

Start and enable the FTP service

Verify the service is running

Confirm FTP is listening on port 21

```
root ~  
🐧 lsof -i:21  
COMMAND  PID USER  FD  TYPE DEVICE SIZE/OFF NODE NAME  
vsftpd    4492 root   3u  IPv6  25212      0t0  TCP *:ftp (LISTEN)
```

Q2. Configure firewall for FTP access

Check if the FTP service is available in the firewall services

Allow FTP service to run permanently

Reload firewall configuration

Verify FTP service is allowed

```
root ~  
🐧 firewall-cmd --add-service=ftp --permanent  
success  
root ~  
🐧 firewall-cmd --reload  
success  
root ~  
🐧 firewall-cmd --list-all  
public (default, active)  
  target: default  
  ingress-priority: 0  
  egress-priority: 0  
  icmp-block-inversion: no  
  interfaces: ens160  
  sources:  
  services: cockpit dhcpv6-client ftp http https ssh  
  ports:  
  protocols:  
  forward: yes  
  masquerade: no  
  forward-ports:  
  source-ports:  
  icmp-blocks:  
  rich rules:
```

Q3. Configure anonymous FTP access

Place a file named public.txt in the FTP public directory

Enable anonymous login in the configuration

Restart FTP service

Verify that anonymous users can access and download the file

Q4. Configure local user authentication

Create a local user ftpuser

Set password for the user

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Download multiple files from server

```
root@localhost:~# ftp 192.168.220.130
Connected to 192.168.220.130 (192.168.220.130).
220 (vsFTPd 3.0.5)
Name (192.168.220.130:root): ftpUser
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
227 Entering Passive Mode (192,168,220,130,248,203).
150 Here comes the directory listing.
226 Directory send OK.
ftp> pwd
257 "/home/ftpUser" is the current directory
ftp> ls
227 Entering Passive Mode (192,168,220,130,81,196).
150 Here comes the directory listing.
226 Directory send OK.
ftp> get filename
local: filename remote: filename
227 Entering Passive Mode (192,168,220,130,131,72).
550 Failed to open file.
ftp> put filename
local: filename remote: filename
local: filename: No such file or directory
ftp> mget *
ftp> mput *
mput Desktop? y
Desktop: not a plain file.
mput Documents? n
```

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Q6. Restrict user to home directory

Configure FTP to restrict users to their home directory

Restart FTP service

Verify user cannot access directories outside home

Q7. Disable anonymous access

Modify configuration to disable anonymous login

Restart FTP service

Verify anonymous login is denied

Q8. Configure custom upload directory

Create a directory for uploads

Set appropriate permissions for writing

Enable upload capability in FTP configuration

Upload files from client to this directory

Q9. Troubleshoot FTP connection issues

Verify FTP service status

Check if FTP port is active

Verify firewall configuration

Restart FTP service if required

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Verify full working setup